Raj Reddy

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Education

University of Utah — **Salt Lake City, UT**
B.S. Computer Science — *August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies — **Dallas, TX**
Associate Software Engineer — *June 2025 – Present*

- Developed signal processing modules in Python and C++ on embedded systems, collaborating with hardware teams to optimize performance and resource utilization.
- Built microservices for high-frequency data ingestion, processing, and storage, leveraging Podman and serverless functions.
- Automated CI/CD testing with pipelines, integrating static code analysis and integration testing.

Doxy.me — **Charleston, SC**
Software Engineer — *August 2024 – May 2025*

- Built a real-time mental health inference system for Doxy.me's telehealth platform, fine-tuning Llama models to predict depression risk, anxiety, and mood indicators from tone, voice, and facial cues captured over live WebRTC sessions, served via SageMaker and Fargate.
- Improved prediction accuracy of the mental health inference system by applying RL feedback loops over structured clinical signals and physician-validated outcomes from live video consultations.
- Shortened the gap between model output and clinician action by developing TypeScript frontend components embedded within the video consultation interface, surfacing AI-generated insights directly at the point of care.
- Eliminated environment drift and enabled zero-downtime releases by standardizing CI/CD configuration with YAML-based pipelines across the ML stack.

University of Utah — **Salt Lake City, UT**
Undergraduate Research Assistant – FuTURES Lab — *May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

HEXstream — **Chicago, IL**
Software Engineering Intern — *May 2022 – Aug 2022*

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.
- Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.

Projects

University Course Planning/Review Platform *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Ray Tracing Engine: Created an interactive WebGL-based ray tracing engine in JavaScript, featuring realistic reflections, dynamic lighting, and customizable environment maps. Implemented shaders and user controls for rendering techniques, scene adjustments, and interactivity.